

Building IOT Applications



Introduction

- Security means preventing safeguarding connected device and network in the internet of things (IoT) falling in the hands of unauthorized people.
- The sender of data may wish to be assured that only the intended party actually receives designated data.
- The receiver must authenticate first that is the one who is authorized to receive data and thereafter the sender exchanges data.



Arduino

- Arduino is an open source electronics platform accompanied with hardware and software to design, develop and test complex electronics prototypes and projects.
- Arduino was created in the year 2005 by two Italian engineers **David Cuartielles** and **Massimo Banzi** with the goal of keeping.



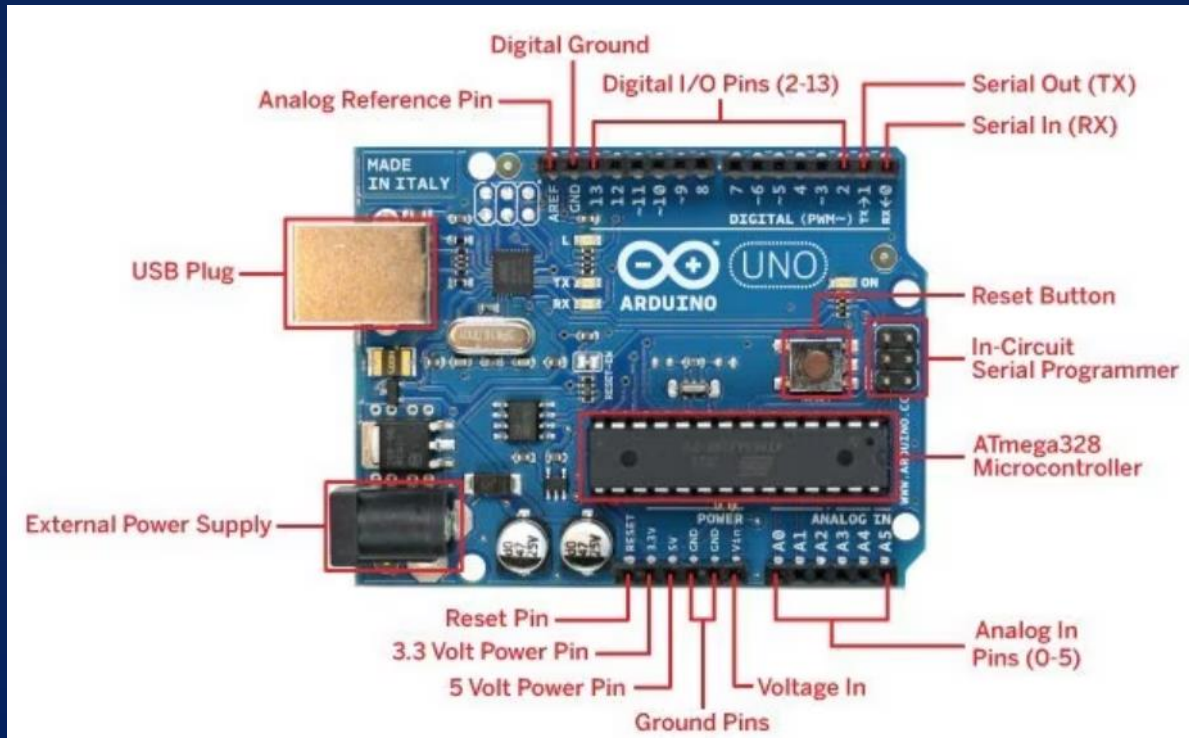
Type of Arduino Board

- Arduino Uno
- Arduino Leonardo
- Arduino 101
- Arduino ESPLORA
- Arduino Mini/Mini Lite
- Arduino Micro
- Arduino Nano
- Arduino Mega
- Arduino Zero
- Arduino M0 PRO
- Arduino Yun
- Arduino industrial 101
- Arduino Tian
- Lilypad Arduino USB
- Lilypad Arduino main board



Arduino Uno

- The Arduino Uno is one kind of microcontroller board based on ATmega328, and uno is an Italian term which means one.



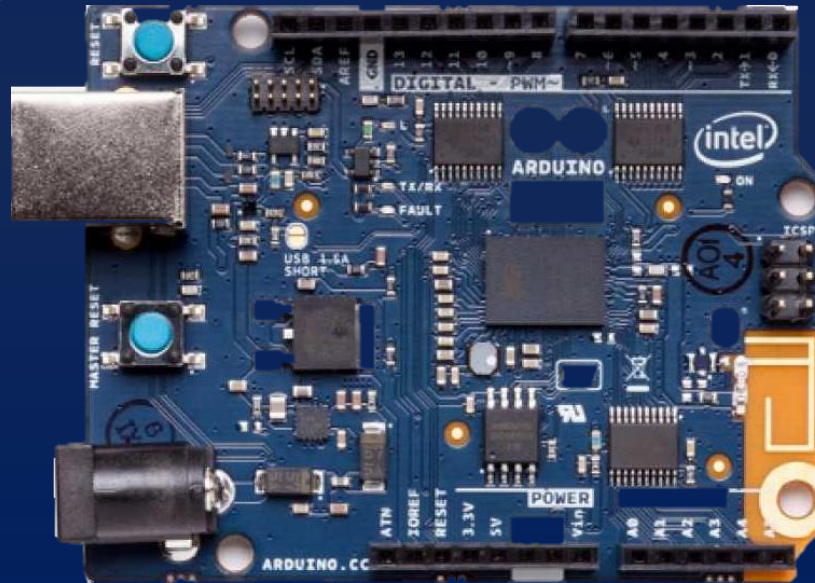
Arduino Leonardo

- Arduino Leonardo is another entry-level board that uses the ATmega32u4 Microcontroller.
- It is Arduino's First development board to use one microcontroller with built-in USB.



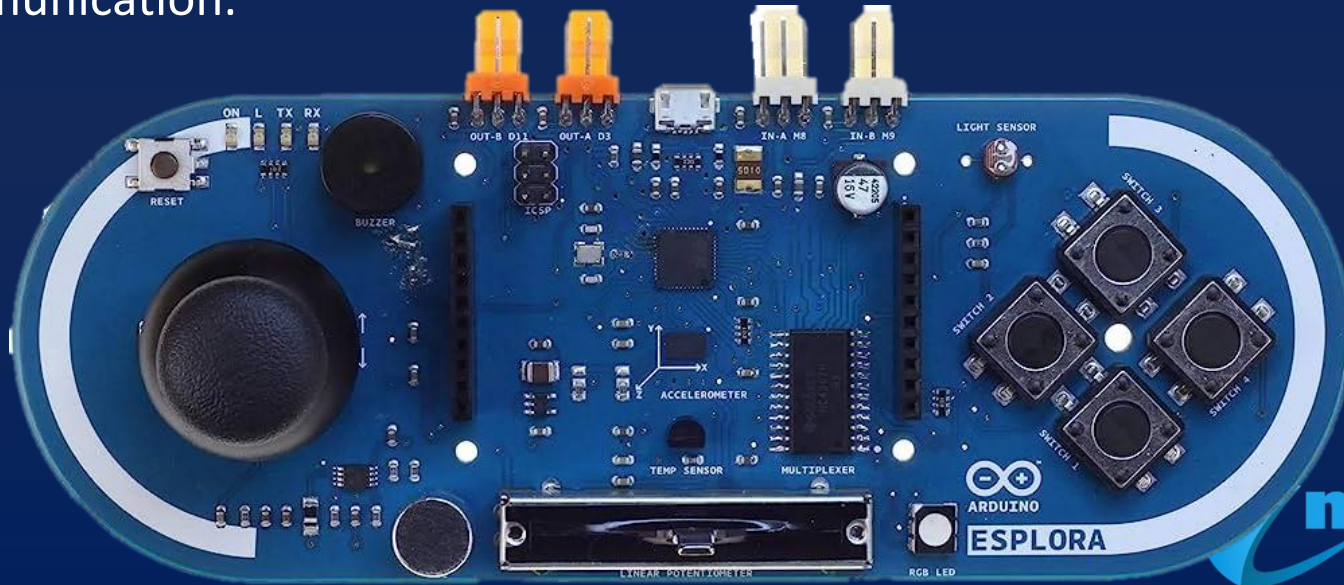
Arduino 101

- The Arduino 101 is simple and has 32-bit Intel Curie microcontroller.
- It's a board that combines the universal appeal of Arduino with the latest technologies- like the Intel Curie module, Bluetooth LE capabilities, and a 6-axis accelerometer.



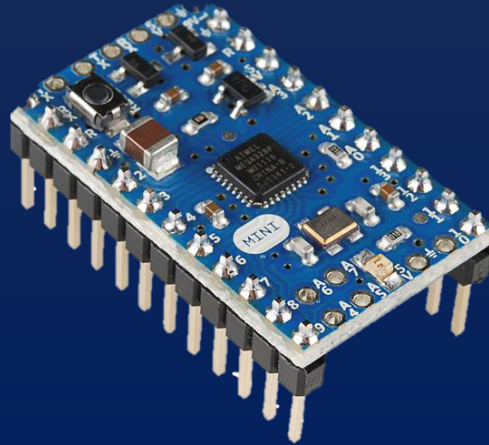
Arduino ESPLORA

- The Arduino esplora is an Arduino Leonardo based board with integrated sensor and actuators.
- It uses the ATmega32u4 microcontroller. The esplora has built-in USB communication.



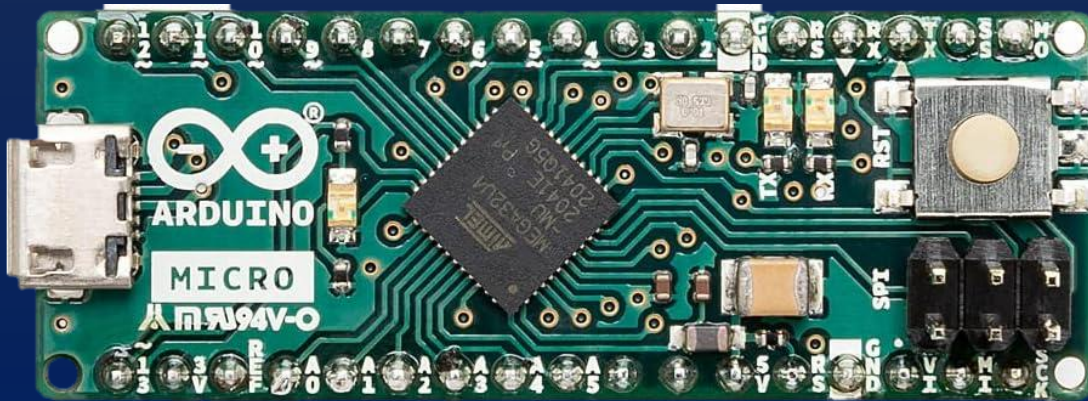
Arduino Mini/Mini Lite

- It is smallest Arduino product which is a 24-pin microcontroller without any connector soldered.
- The unit features 8 analog pins and 14 digital pins.
- The module is based around the ATME-GA168 processor.



Arduino Micro

- Arduino Micro is the smallest board of the family, used for interactive computing.
- The micro comes with a built-in USB which makes connection with a computer.
- This microcontroller board is based on ATmega32U4.



Arduino Nano

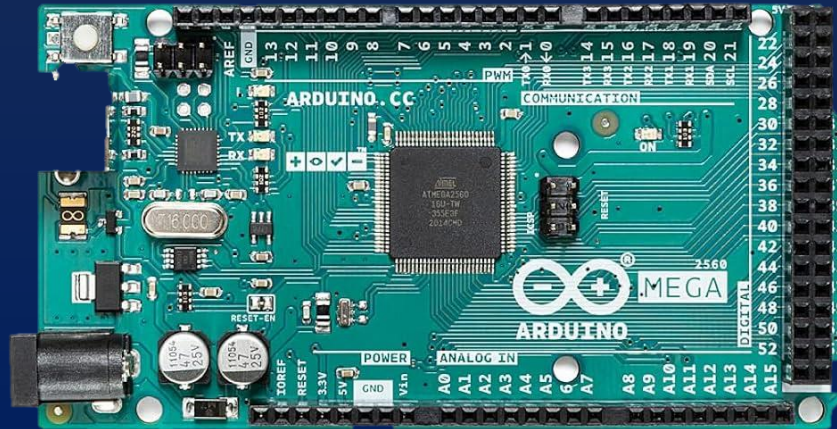
- It is breadboard friendly board based on the ATmega328.
- The Nano was the first mini breadboard-compatible board to have onboard USB.
- This microcontroller distinguishes itself from others by having the USB to serial chip and connector onboard.
- The Nano has 8 analog pins and 14 digital pins.



max

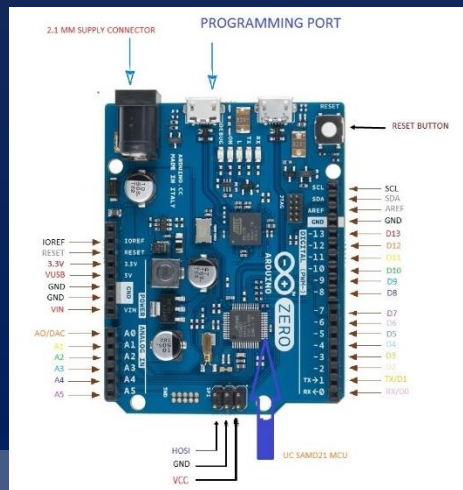
Arduino Mega

- It is based on **ATmega2560** microcontroller.
- It has more digital input/output pins, analog inputs, a USB connection, a power jack, and a reset button.
- The large number of pins make this board very handy for project that require a bunch of digital inputs or outputs (like lots of LEDs or buttons).



Arduino Zero

- Arduino Zero provides 32-bit extension to the platform established by Arduino Uno R3.
- It uses the **ATSAMD21G18** microcontroller.
- Unlike most Arduino and Genuino boards, the Zero runs at **3.3V**.
- Applying voltage higher than 3.3V to any I/O pin could damage the board.



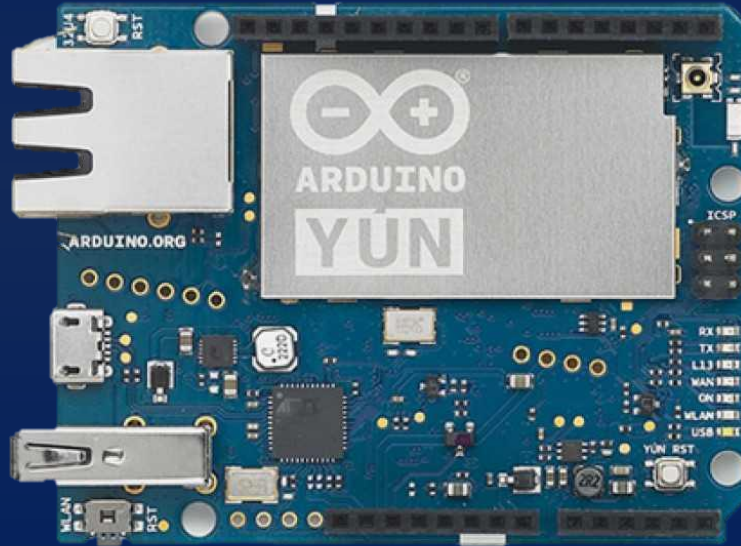
Arduino M0 PRO

- The Arduino M0 Pro board equipped with **Atmel Embedded Debugger (EDBG) or in-board debugging.**
- It is based on **ATSAMD21G18** microcontroller.



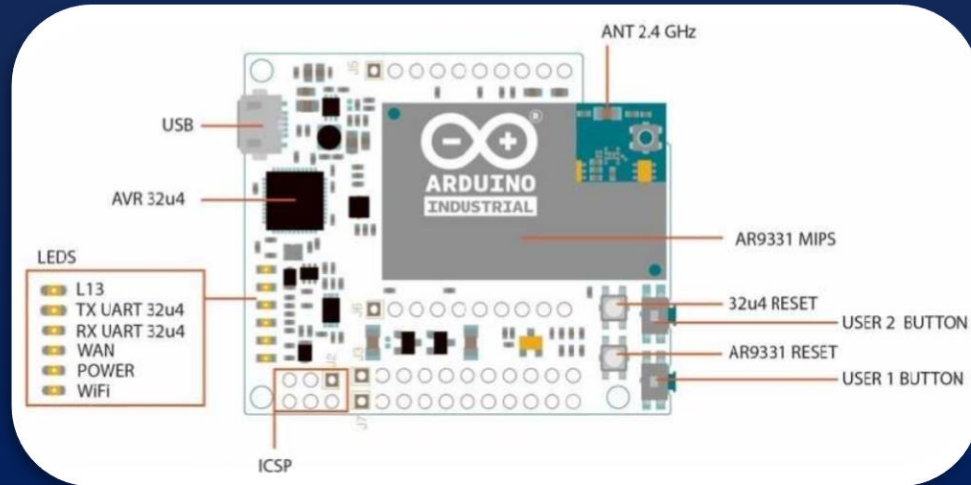
Arduino Yun

- The Arduino Yun uses a **ATMega32U4** chip instead of the ATmega328.
- This extra chip allows this board to **run Linux in addition to the Arduino operating System.**



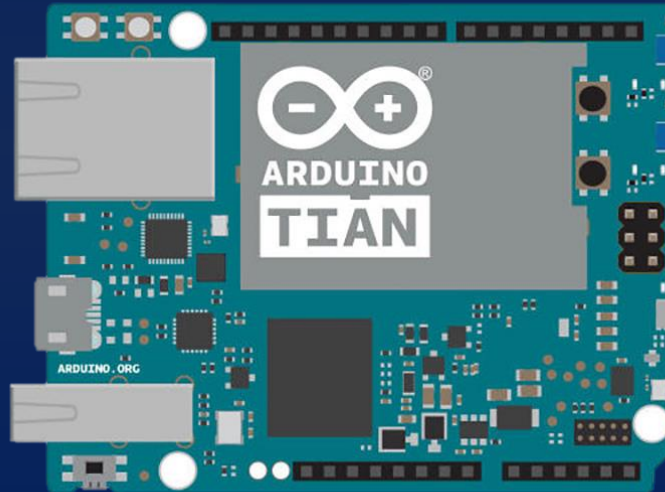
Arduino Industrial 101

- Arduino Industrial 101 is an evaluation board for **Arduino 101 LGA module**.
- The **ATmega32u4** microcontroller is **integrated in the baseboard**.
- The Arduino Industrial 101 has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers.



Arduino Tian

- The Arduino tian unleashes the power of small computer running Linux connected to a 32-bit microcontroller.



Lilypad Arduino USB



Lilypad Arduino main board

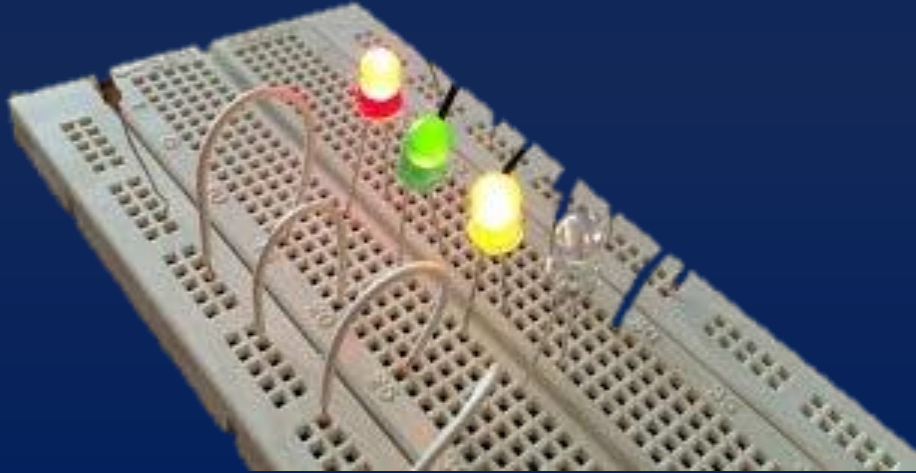


Arduino clones



Breadboard

- A breadboard is circuit board that is used to make temporary circuit and prototyping.
- Prototyping is the process of testing out an idea by creating a preliminary model from which other forms are developed or copied.



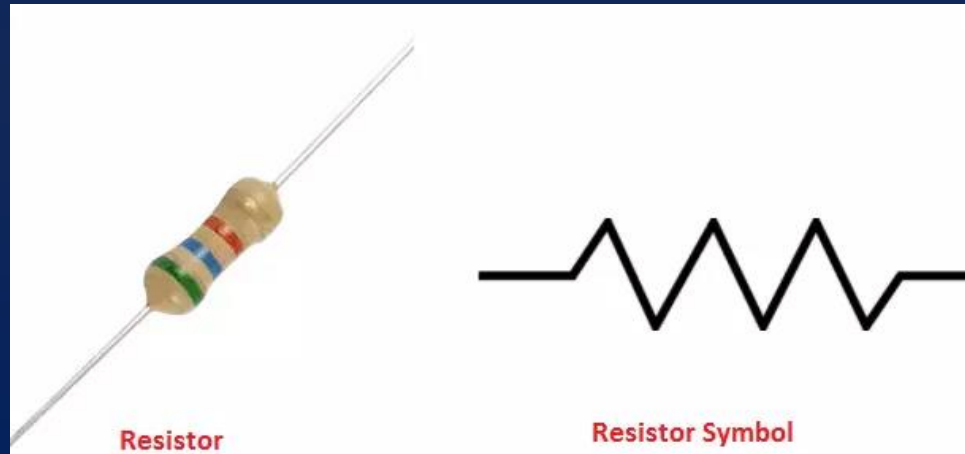
LED (Light Emitting Diode)

- LED (Light Emitting Diode) is one of the most commonly used source of light now-a-days.
- They do many different jobs in all kind of devices. For instance, they are widely used for displaying the time and many other type of data on screen in certain display devices.



Resistor

- A resistor is an electronic component that limits or regulates the flow of electronic current in an electronic circuit.
- Resistor can also be used to provide a specific voltage for an active device such as transistor.



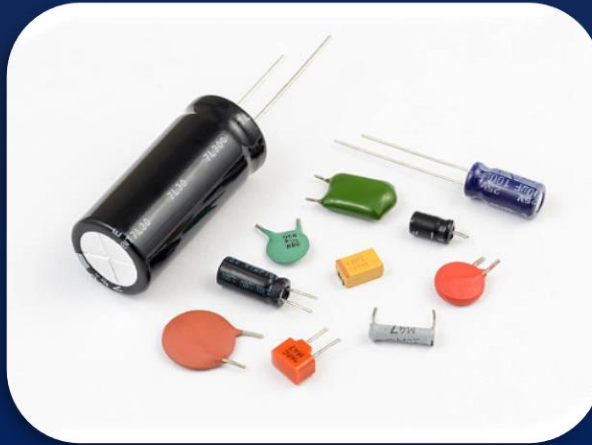
Jumper wires

- Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering.
- Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed.



Capacitor

- A capacitor is a device that stores electrical energy in an electric field.
- It is a passive electronic component with two terminals.
- The effect of capacitor is known as **capacitance**.



Writing Arduino Code

- The program in Arduino consist of two function. These are:
- Void setup()
- Void loop()



Void Setup()

- This is the first routine that begins when the Arduino starts functioning.
- Every Arduino sketch must have a setup() function.
- This function is executed only once throughout the entire program functioning.
- The setup function contains the initialization of every pin you intend use in your project for input and output.
- It is also used to initialize class



THANK YOU

Address: G. R. Complex Preetam Nagar, Dhoomanganj
Prayagraj Uttar Pradesh 211011

Mobile No.: 9598948810, 8874588766, 9532878816

Website : www.infomaxacademy.com

